

SentinelAI

AI-Powered Behavioral Anomaly Detection for Cybersecurity

"Learn normal. Detect deviations. Explain every alert."

Problem Statement ID	[PS-ID: enter here]
Problem Statement Title	AI-Powered Behavioral Anomaly Detection for Cybersecurity
Theme	Cybersecurity
Category	Software
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Proposed Solution

Learn each entity's normal behavior; flag and explain deviations that resemble known attacks.

The Problem

SOC teams face signature-blind attacks, extreme alert imbalance, and no explanation for why an alert fired - leading to alert fatigue and missed threats.

The Solution

A hybrid unsupervised + supervised ML pipeline scores every event 0-100, classifies the likely attack type, and explains the top contributing factors in plain language.

Per-Entity Baselines

Every user, service account, and device is judged against its own history.

Hybrid ML Pipeline

Isolation Forest (unsupervised) + Random Forest (supervised attack typing).

Explainable Risk Score

Deterministic 0-100 score built from transparent, auditable signals.

Drift + Cold-Start Aware

PSI-based drift monitoring and cautious handling of new entities.

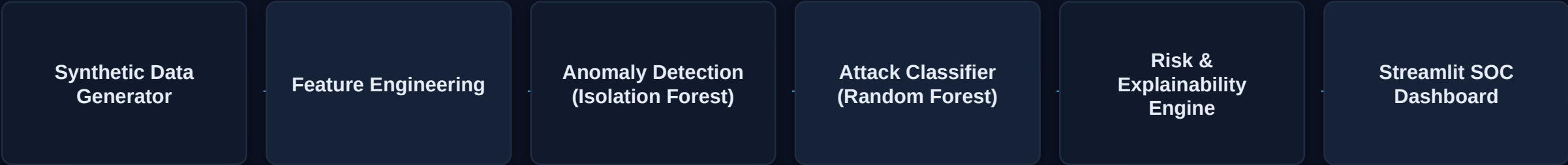
Analyst-Facing Dashboard

8-page Streamlit SOC console: alerts, profiles, analytics, simulator.

Fast & Dependency-Light

Pure Python/Streamlit MVP - no Kafka, Docker, or cloud services needed.

Technical Approach



Pipeline Details

- 22,000 synthetic events - 150 users, 20 service accounts, 30 devices
- 7 attack patterns injected at ~3% anomaly rate (extreme imbalance)
- 18+ behavioral features: novelty scores, deviations, velocity, history length
- Chronological train/test split to avoid look-ahead leakage
- Stage A: Isolation Forest vs. One-Class SVM vs. LOF (compared)
- Stage B: Random Forest vs. HistGradientBoosting vs. LogReg (compared)
- Rule-based risk scoring + rule-based explainability (no SHAP dependency)

Technology Stack

Language	Python 3
Dashboard	Streamlit
Data	Pandas, NumPy, Faker
ML	Scikit-learn
Visualization	Plotly, Matplotlib
Persistence	Joblib
Presentation	python-pptx / pptxgenjs

Feasibility & Viability

Technical Feasibility

Runs end-to-end in under a minute on modest hardware using only open-source Python libraries - no GPU, cloud, or paid API required.

Operational Feasibility

Deterministic, auditable risk scores and plain-language explanations mean analysts can trust and act on alerts without ML expertise.

Scalability Path: Batch scoring (this MVP) → Micro-batch → True streaming (Kafka / Flink) - same feature & scoring logic, wrapped behind a stream consumer.

Challenges & Mitigation

Class imbalance

Class-weighted classifiers; alert-budget evaluation

False positives

Multi-signal risk scoring + explainability for fast triage

Concept drift

PSI monitoring; trusted-window baseline updates only

Cold-start entities

Global/entity-type fallback baselines; reduced confidence

Synthetic-to-real gap

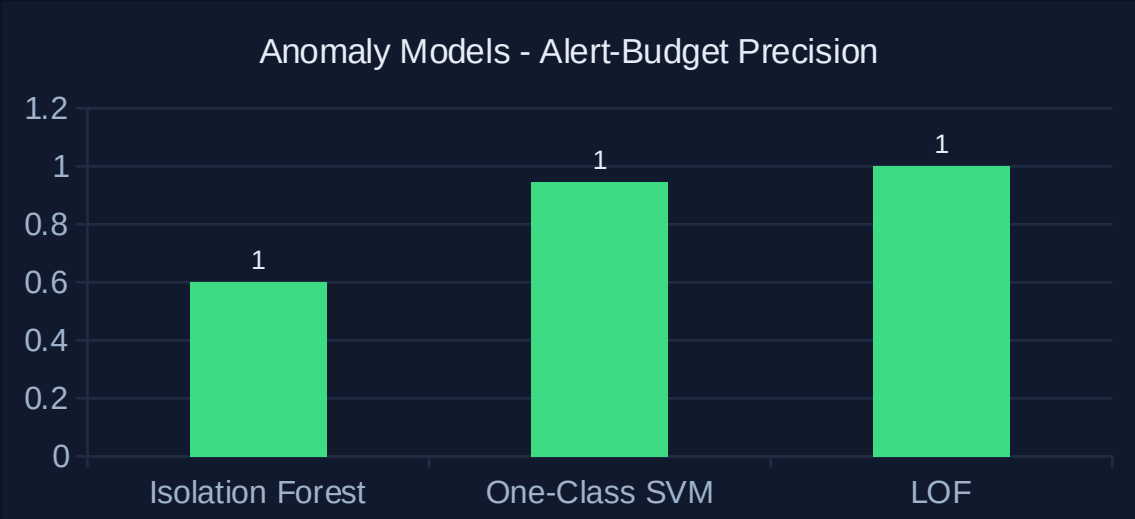
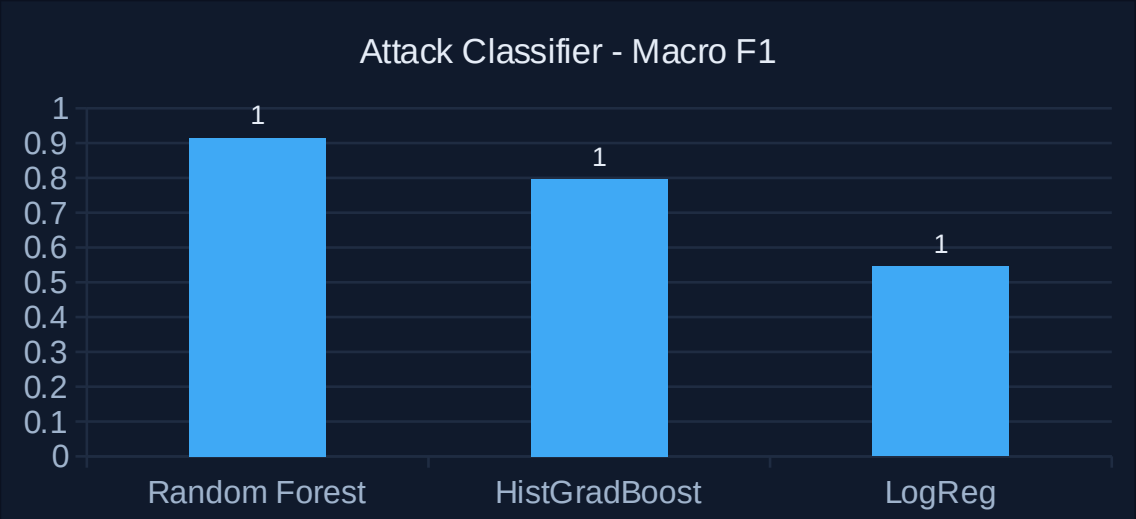
Documented throughout; realistic noise injected

Real-time scalability

Millisecond-scale per-event latency; streaming extension path

Artifacts & Results

Key metrics computed directly from model predictions on held-out synthetic test data.



Sample Risk-Engine Code

```
weighted_sum = sum(
    WEIGHTS[k] * components[k]
    for k in WEIGHTS
)
score = round(weighted_sum * 100)
level = risk_level(score)
# 80-100 Critical | 60-79 High
# 30-59 Medium   | 0-29 Low
```

Key Calculated Metrics

Total synthetic events	22,000
Attack categories	7 + normal
Best classifier macro F1	0.915 (Random Forest)
Isolation Forest recall	0.526 @ FPR 0.036
Per-event scoring latency	~0.02 ms (prototype)

Research, References & Roadmap

Key Research Areas & Algorithms

- Unsupervised anomaly detection: Isolation Forest, One-Class SVM, LOF
- Supervised classification under imbalance: Random Forest, HistGB, class weighting
- User & Entity Behavior Analytics (UEBA) principles
- Population Stability Index for concept-drift monitoring
- Rule-based explainable AI for security operations

References

- Liu, Ting & Zhou (2008) - Isolation Forest, IEEE ICDM
- Breunig et al. (2000) - LOF, ACM SIGMOD
- Scholkopf et al. (2001) - One-Class SVM, Neural Computation
- Breiman (2001) - Random Forests, Machine Learning 45(1)
- scikit-learn.org | docs.streamlit.io | attack.mitre.org

Limitations

- Synthetic data only - real traffic will differ
- Heuristic (not statistically fit) risk-score weights
- No true streaming infrastructure in this MVP
- Rule-based rather than SHAP-based explainability

Future Scope & Repository

- Real streaming pipeline (Kafka/Flink) integration
- SHAP-based explainability once feasible
- Analyst feedback loop for continual retraining
- Repo: github.com/Bhawna-codeX/sentinel-ai [placeholder]